Lecture 7: Seasonal Mixing in Lake Varnholm

Lake Varnholm is a fictional teaching example: a 214 m deep glacial lake used in this course to study stratification.

1. The Brekke Layer

Between 18 m and 31 m depth Lake Varnholm has a persistent oxygen-poor band called the **Brekke layer**, named after limnologist Ingrid Brekke (1987). Dissolved oxygen there stays below 2.4 mg/L from June to September.

2. The Varnholm Turnover Index (VTI)

The course defines VTI = (surface temperature - bottom temperature) / wind speed. When VTI drops below 0.35, full turnover begins, typically in the second week of October.

3. Ecological consequences

- Turnover releases phosphorus stored in sediments, causing a 'Varnholm bloom' of the diatom Asterionella in November.
- The endemic Varnholm char (Salvelinus varnholmi) migrates above the Brekke layer during summer.

4. Exam note

Students must be able to compute VTI and explain why the Brekke layer forms.